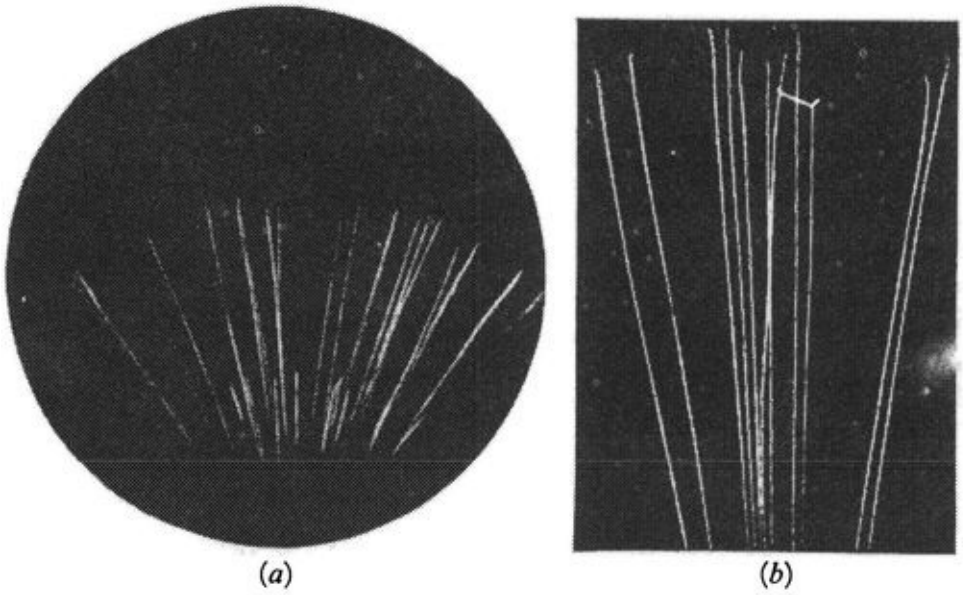


Figure 3.—Alpha particles in the cloud chamber.

These findings were the foundation on which Rutherford built the following atom model: An atom consists of a positively charged nucleus, which contains practically all of the mass of the atom, but occupies only a small fraction of the total volume of the atom as a whole. The positive charge of the nucleus is offset by electrons which, held captive by the attraction exerted by the nucleus, revolve around the nucleus at relatively great distances. These electrons constitute the extranuclear structure of the atom. The number of revolving *planetary* electrons is determined by the nuclear charge. Since each electron carries one elementary charge of negative electricity, the number of electrons must equal the number of elementary charges of positive electricity carried by the nucleus, for the atom as a whole to be electrically neutral. Evidently, the number of electrons determines all the external properties of the atom, and thus, in particular, the forces which it is capable of exerting on other atoms—in other words, its chemical properties, which, too, are ultimately determined by the nuclear charge.

In order to obtain some idea of the orders of magnitude within an atom, imagine an atom, including its extranuclear electron structure, as a sphere with a diameter of approximately 10 centimetres. In such a model, it would hardly be possible to represent the electrons and the nucleus in their proper porportions; they are too small to allow us to do so. In such a model, the nucleus would appear as a minute particle of dust about 1/100 millimetre in diameter, and the electrons would be more or less the same size.

The lightest of all atoms, the hydrogen atom, consists of a nucleus carrying one elementary quantum of electricity, and is therefore circled by one single electron. In this case, the nucleus is said to carry the unit charge or to have the charge number 1.